

Gaurav Singh

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Professional profile

Software developer at Pay Advantage building payment systems with C#, .NET, TypeScript, and Vue. Research background in applied machine learning and mathematical modelling; previously headed a postgraduate design-computation programme and founded the Experimental Maths Lab.

Technical skills

Software: C#, .NET, TypeScript, Vue, REST APIs, SQL, Git, and Azure DevOps.

Machine learning and modelling: Python, scikit-learn, computer vision, time-series and generative modelling, statistical inference, and nonlinear optimisation.

Experience

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| 2025-present | Software Developer, Pay Advantage, Australia <ul style="list-style-type: none">◦ Develop payment-platform software covering recurring billing, direct debit, subscriptions, bank-account runs, card payments, hosted payment pages, and API integrations.◦ Work with C#, .NET, TypeScript, Vue, Azure DevOps, and REST APIs. |
| 2017-2022 | Faculty Member and Researcher, Srishti Manipal Institute , Bangalore, India <ul style="list-style-type: none">◦ Headed the Master of Design in Design Computation.◦ Founded the Experimental Maths Lab.◦ Taught programming, algorithms, machine learning, physical computing, and interaction design. |
| 2018-2022 | Design Copilot (Part-time), Topcoder, Remote <p>Ran design challenges for Fortune 500 clients and guided distributed groups of designers from brief interpretation through final delivery, after several years competing on the platform.</p> |

Selected projects

AATP: agricultural traceability demonstrator. Led development and built the complete software demonstrator during my master's at Griffith University, supervised by the team at Anonymo Labs. Used VON Network, ACA-Py and Python/Flask to integrate digital identity wallets, credential schemas and a web interface for issuing verifiable credentials and tracing products through production, packaging and transport.

Machine learning for auxetic deformation. Co-developed a workflow combining finite-element simulation data, K-means image labelling and logistic regression to automate deformation-mode detection in re-entrant honeycomb structures. The study detected four of six modes; published in *International Journal of Protective Structures* (2025).

Heavy-transport decarbonisation modelling. Co-developed a nonlinear framework linking Australian emissions accounting with fleet-transition economics. Evaluated lifecycle emissions, ownership costs and return on investment across 270 scenarios over a 25-year horizon; published in *Fractals* (2026).

Education

2023-2025	Master of Information Technology Griffith University, Gold Coast, Australia; GPA 6.31/7 Thesis: "Enhancing Climate Forecasting with Advanced Generative Models for Synthetic AQI Data", supervised by Dr. Elizabeth Chang.
2013-2015	Advanced Diploma, Innovation & Experience Design <u>Srishti Institute of Art, Design and Technology, Bangalore, India</u>
2008-2012	B.Tech, Computer Science & Engineering JNTU Anantapur, India; First Class with Distinction

Selected publications

A Framework for Visual-Mathematical Literacy in Applied Machine Learning: Why Representation Choices Shape What Models Can Learn. G. Singh, R. S. Dhari. Machine Learning: Engineering, 2026. <https://doi.org/10.1088/3049-4761/ae7df3>

Techno-Economic Pathways Modeling and Nonlinear Optimized SEEA-ROI Longitudinal Dynamic Simulation for Decarbonizing Australian Heavy Transportation Systems. G. Singh, E. Chang, Y. Karaca. Fractals, 2026. <https://doi.org/10.1142/S0218348X26400633>

Automated detection of deformation mechanisms in re-entrant honeycomb auxetics using machine learning. G. Singh, R. S. Dhari, Z. Javanbakht. International Journal of Protective Structures, 2025. <https://doi.org/10.1177/20414196241281069>

Selected recognition

2024	Griffith Award for Academic Excellence (Griffith University)
2024	STEM Professional Gold Achievement (Griffith University)
2017	Topcoder Open Design Champion (India Regionals)

Portfolio: <https://gaurav-singh.info> · [Full academic CV](#)